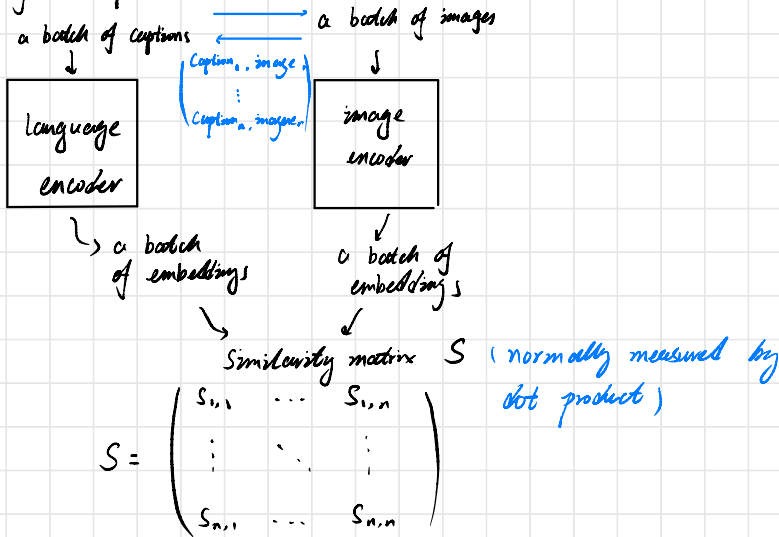


1. Contrastive Learning

Similar things should be grouped together,
and dissimilar things should be pushed apart.

Structure for CLIP



Loss function (Symmetric cross-entropy loss)

$$L_i^{(I \rightarrow T)} = -\log \frac{\exp(s_{i,i}/\tau)}{\sum_{j=1}^n \exp(s_{i,j}/\tau)}$$

→ this part can be interpreted
as probability (the probability of
image i to find the correct text)

$$L_i^{(T \rightarrow I)} = -\log \frac{\exp(s_{i,i}/\tau)}{\sum_{j=1}^n \exp(s_{j,i}/\tau)}$$

final loss:

$$\sum_{i=1}^n \frac{(L_i^{(I \rightarrow T)} + L_i^{(T \rightarrow I)})}{2n}$$

Other methods:

pairwise Contrastive

look at a pair once, Some pairs are positive,

Some pairs are negative

(image i , caption i)

(image i , caption j)

$$\text{loss} = \gamma_i D^2 + \underbrace{(1 - \gamma_i) \cdot \max(0, m - D)}_{\text{if the pair is positive, minimize } D}$$

if the pair is negative
push the distance
to more than m

minimize D

triplet loss

look at three data points at a time

1. Anchor (a): the baseline image

2. Positive (p): paired caption

3. Negative (n): a unpaired caption

$$\text{loss} : \max(0, \underline{d(a, p)} - d(a, n) + m)$$

the distance of a and p
should be smaller than a to n
by at least m .